PNEUMONIA DETECTION FROM CHEST X-RAY IMAGES USING CONVOLUTIONAL NEURAL NETWORKS

Cynthia Du
University of California, Los Angeles
UID 605915782
cynthiadu@ucla.edu

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ABSTRACT

The dataset used is Chest X-Ray Images (Pneumonia) on Kaggle, which is available to the public and is found here: https://www.kaggle.com/datasets/paultimothymooney/chest-xray-pneumonia.

This paper presents a deep learning investigation into automated pneumonia detection from chest X-ray images. Using a dataset of 5,856 labeled pediatric x-rays, we first trained a custom CNN, then stress-tested it to identify its limitations. Those findings motivated a second approach using ResNet50 with transfer learning and an enriched augmentation pipeline. The custom CNN achieved 80.29% accuracy on mirrored test images with 97.44% pneumonia recall (true positive rate); the fine-tuned ResNet50 achieved 80.93% accuracy with 98.97% recall (true positive rate). Stress tests — horizontal mirroring and synthetic image degradation — revealed shortcut learning behaviors that directly informed our redesign. Results highlight the clinical trade-off between sensitivity (prioritizing the detection of all actual infections) and specificity (true negative rate) and the value of iterative model development in medical imaging.

1 Introduction and Background

Pneumonia is one of the leading infectious causes of mortality worldwide, and timely diagnosis via chest x-ray is critical for treatment. Automated classification systems powered by Convolutional Neural Networks (CNNs) have attracted substantial research interest as a means of supporting radiologists, particularly in resource-limited settings where expert interpretation may be delayed.

CNNs learn spatial hierarchies of features directly from pixel data: early layers detect edges and textures while deeper layers capture abstract, task-relevant structures. This property makes them naturally suited to medical images, where relevant pathology (e.g., pulmonary consolidation, opacity) is expressed as spatial patterns. This project explores two modeling approaches: a simple custom CNN and the pre-trained ResNet50 residual network. We compare their performance and robustness on this binary classification problem. Beyond classification accuracy, we were also interested in what the models have learned and how fragile those representations are to transformations that would not mislead a trained radiologist.

2 Dataset Description and Exploratory Data Analysis

2.1 Dataset Overview

We used the publicly available Chest X-Ray Images (Pneumonia) dataset hosted on Kaggle, originally curated by researchers at the Guangzhou Women and Children's Medical Center. The dataset contains 5,856 labeled JPEG chest radiograph images of 128x128 pixels, all of which are from pediatric patients aged one to five years. Each image is labeled as either NORMAL (healthy) or PNEUMONIA (infected). The dataset is pre-partitioned into three splits: a

training set of 5,216 images, a validation set of 16 images, and a test set of 624 images. The training set comprises 1,341 Normal and 3,875 Pneumonia images, representing a substantial class imbalance of roughly 1:2.9 in favor of the Pneumonia class. The validation set, with only 8 images per class, is critically small, at less than 0.3% of the total dataset, making it statistically unreliable for meaningful hyperparameter tuning. The test set contains 234 Normal and 390 Pneumonia images.

2.2 Exploratory Data Analysis

Before preprocessing or modeling, we conducted a thorough exploratory data analysis (EDA) to understand the statistical properties of the dataset and to inform subsequent design decisions. We began by visualizing random samples from the training set for both classes, which confirmed the expected clinical appearance: Normal X-rays show clear, well-defined lung fields without opacity, while Pneumonia X-rays frequently display cloudy or hazy regions.

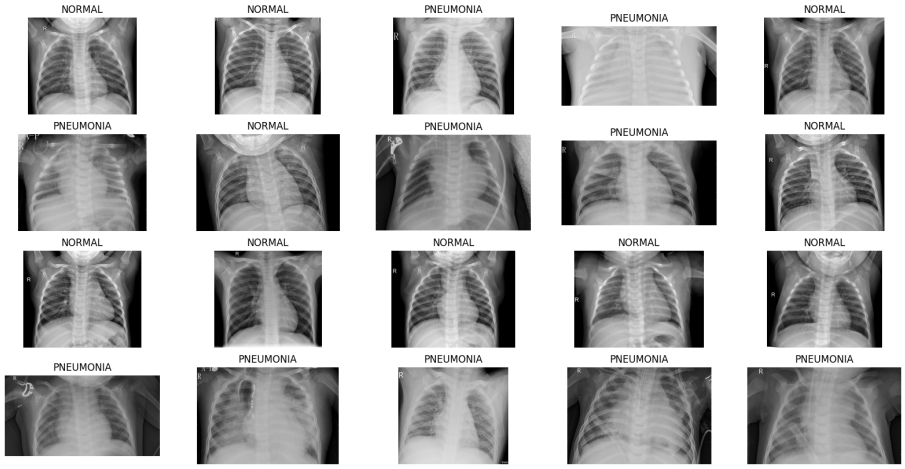
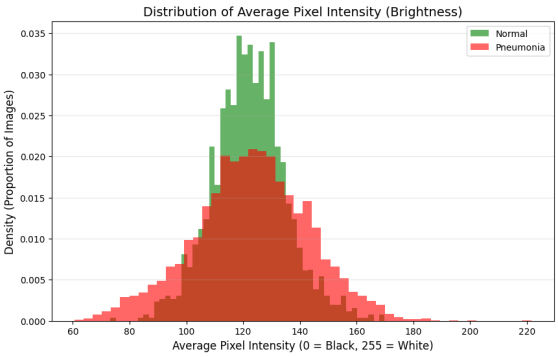
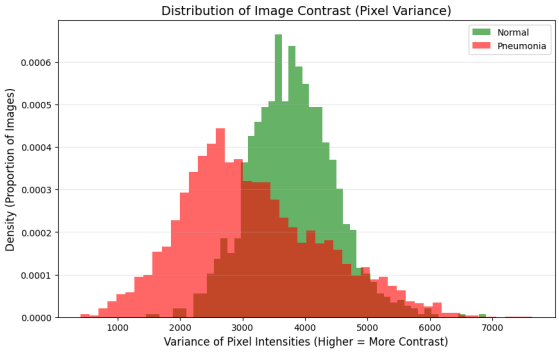


Figure 1: Sample Chest X-rays

We then computed two key image statistics for every image in the training set: mean pixel intensity (a proxy for overall brightness) and standard deviation of pixel intensity (a proxy for contrast or textural complexity). These were computed directly from the raw grayscale pixel arrays prior to normalization. Mean pixel intensities were broadly similar between Normal and Pneumonia images, suggesting overall brightness alone is not a reliable discriminating feature. However, contrast (pixel variance) revealed a more meaningful difference: Normal X-rays had a markedly higher mean contrast compared to Pneumonia images. This finding makes sense clinically. Healthy lungs are essentially bags of empty air, which show up as pure black on an X-ray, contrasting sharply against white ribs and the heart. When a patient has pneumonia, their lungs fill with fluid and pus. This creates cloudy, gray opacities. Mathematically, this reduces the variance of the image because the darkest parts of the lung get lighter, pushing everything closer to a uniform gray.



(a) Pixel Brightness Distribution by Class



(b) Image Contrast Distribution by Class

Figure 2: EDA of brightness and contrast statistics for the training data

Finally, we plotted the class distribution across all three data splits using bar charts. This visualization confirmed the class imbalance in the training set and highlighted the extremely small validation set, both issues that required explicit preprocessing interventions before model training.

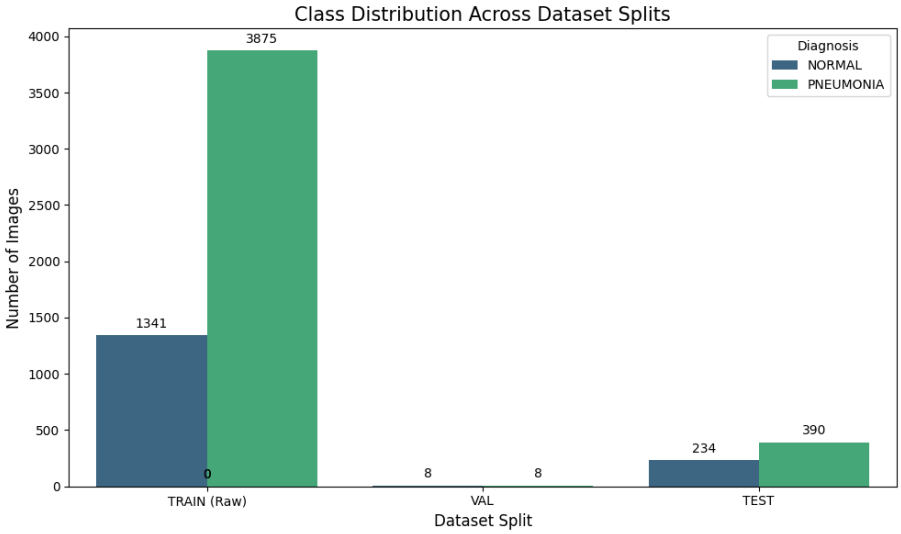


Figure 3: Dataset Distribution.

3 Data Preprocessing

3.1 Fixing Data Imbalance

Training a deep convolutional neural network on heavily skewed data frequently results in majority-class bias. Under the initial conditions, a model is likely to artificially inflate its overall accuracy by simply over-predicting the dominant Pneumonia class, thus failing to learn the subtle, defining features of a healthy lung. Additionally, a validation set of only 16 images cannot effectively trigger Early Stopping mechanisms or accurately reflect the model's generalization capabilities.

To correct these structural issues and establish a rigorous training environment, two primary preprocessing steps were implemented:

- **Stratified Re-splitting:** The original training and validation sets were merged and subsequently re-partitioned using a 15% stratified split. This generated a mathematically robust validation set that accurately reflects the true population distribution, allowing for a much more reliable assessment of model generalization during the training phase.

- **Random Undersampling:** To eliminate the bias in the newly formed training set, a random undersampling technique was applied. The majority class (Pneumonia) was randomly sampled down to match the exact count of the minority class (Normal). This established a perfectly balanced 50/50 training distribution with 1,147 images per class, forcing the network's loss function to optimize based on actual features of the image rather than relying on prior class probabilities.

  Crucially, while the training data was artificially balanced, the validation and test sets were intentionally left in their naturally imbalanced state to ensure that our final performance metrics accurately reflect the model's predictive capabilities in a real-world clinical environment.

  Both the custom CNN and the ResNet50 model were trained on this same balanced, re-split dataset. The only difference in preprocessing between the two pipelines is that the ResNet50 additionally applied on-the-fly data augmentation (described in Section 6) and dynamically resized images to 224×224 to match its expected input dimensions.

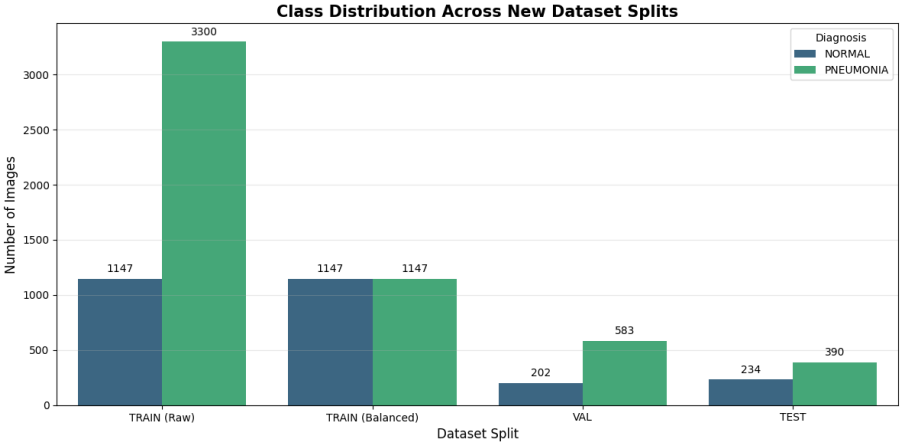


Figure 4: Balanced Dataset Distribution.

4 Custom CNN: Preprocessing, Architecture, and Results

4.1 Preprocessing

Images were loaded in grayscale, kept at 128×128 pixels, and normalized to [0, 1]. Since the CNN architecture expects three-channel input, the single grayscale channel was replicated across three channels via NumPy stacking, producing (128, 128, 3) tensors.

4.2 Architecture

The custom CNN was built using Keras's Sequential API: two convolutional blocks (Conv2D → Batch Normalization → MaxPooling2D) with 32 and 64 filters respectively, a Flatten layer, a Dense(128, ReLU) layer with Dropout(0.5), and a Dense(1, sigmoid) output. The first block detects low-level features like edges and textures; the second captures higher-level lung structures. Batch Normalization stabilizes activations and speeds convergence; Dropout prevents overfitting by randomly zeroing 50% of activations during training. The model was compiled with Adam (lr=0.0001) and binary cross-entropy loss, trained for 10 epochs with batch size 32.

4.3 Evaluation Metrics

Standard accuracy is a flawed metric for medical models because it treats all errors equally. To properly evaluate clinical safety, we focused on confusion matrices and ROC AUC scores. The confusion matrix let us specifically track False Negatives, which is the worst-case scenario where a pneumonia patient is misdiagnosed as healthy. Minimizing these False Negatives allowed us to maximize Recall (true positives). Furthermore, the ROC AUC score confirmed the model's overall strength. It displays that the network genuinely learned to separate healthy and infected X-rays across all possible decision thresholds, rather than just relying on a rigid 50% cutoff.

4.4 Results

On the 624-image test set, the custom CNN achieved the following performance:

The model achieved an overall accuracy of 75.80%, with test precision of 72.34% and test recall of 99.23% for the pneumonia class. These results reveal a characteristic pattern: the model is extremely sensitive to Pneumonia (missing only 6 of 390 pneumonia cases, i.e., nearly perfect recall) but has very low specificity for Normal images (correctly identifying only about 37% of normal cases, or approximately 87 out of 234). In practice, the model's decision boundary is biased heavily toward predicting Pneumonia.

The ROC AUC for the custom CNN reflects a model that, while biased, still captures meaningful signal from the data. The high AUC indicates that with different threshold selection, the trade-off between sensitivity and specificity could be adjusted to better suit clinical requirements.

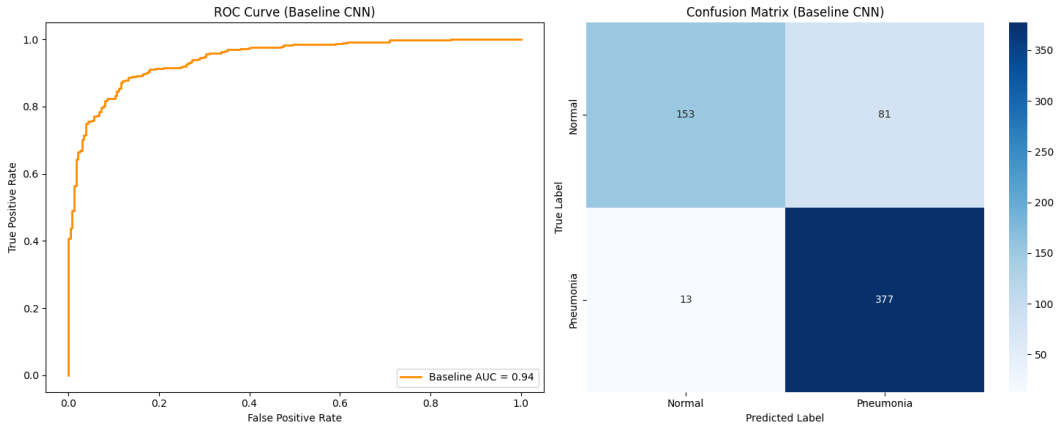


Figure 5: Custom CNN Results.

5 Stress Tests and Robustness Analysis

With a working CNN baseline in hand, we designed two stress tests to probe what the model had actually learned. The goal was to apply image transformations that should not affect diagnostic content, and therefore should not affect a robust model's predictions, in order to identify potential shortcut behaviors before deciding on next steps.

5.1 Horizontal Mirror Test

We horizontally flipped all 624 test images and re-evaluated the CNN against the original ground-truth labels. Since pneumonia is not inherently left-right asymmetric, a robust model should produce similar predictions on mirrored images. On the mirrored test set, the model achieved 80.29% accuracy with 97.44% pneumonia recall and 52% Normal recall.

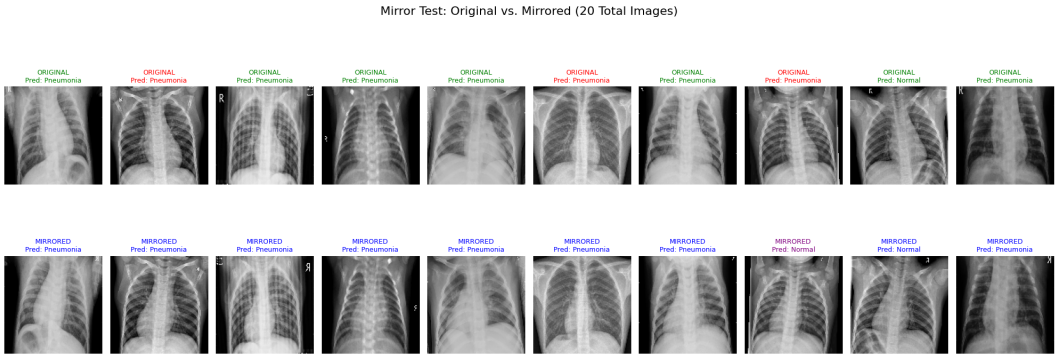


Figure 6: Mirrored Images Test.

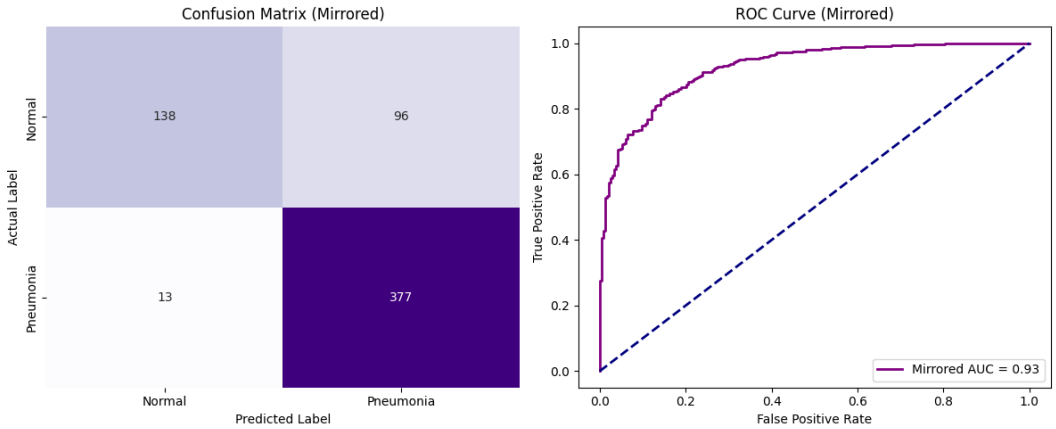


Figure 7: Mirrored Image Results.

Interestingly, accuracy and Normal recall both improved on the mirrored set, suggesting the model has learned some orientation-sensitive features tied to the natural asymmetry of chest anatomy in standard PA-view radiographs (e.g., the cardiac shadow typically falls on the left side of the image). When mirrored, certain Normal images may better match patterns the model associated with Normal during training, temporarily improving specificity. This behavior reveals that the model's decision boundary is partly anchored to spatial position rather than purely to pathological tissue characteristics.

5.2 Cloudy Consolidation Stress Test

To simulate pulmonary consolidation, which is the key radiographic finding in pneumonia, we synthetically degraded all test images by adding uniform haze (random value in [0.1, 0.2] added per pixel) followed by Gaussian blur (5×5 kernel). This increases overall brightness and reduces local contrast, mimicking the appearance of fluid-filled alveolar spaces. The test was designed to determine whether the model had learned genuine structural features or was relying on global contrast as a shortcut.

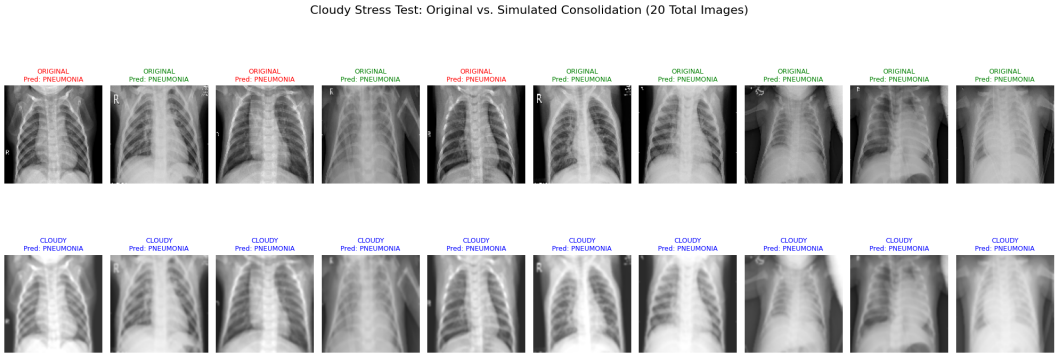


Figure 8: Cloudy Image Test.

The results were highly revealing. Pneumonia recall increased slightly (99.23% → 99.49%), because the haze made all images statistically resemble Pneumonia, pushing the already Pneumonia-biased model to predict Pneumonia even more aggressively. However, Normal recall collapsed from 37% to just 24%, and overall accuracy fell to 71.31%. This confirms that the model relies substantially on global image contrast as a discriminating signal, which is consistent with our EDA finding that Normal images have significantly higher pixel variance. When Normal images are artificially degraded to have Pneumonia-like low-contrast statistics, the model fails to identify them.

Together, the stress tests revealed two key weaknesses: a sensitivity to image orientation suggesting the model relies partly on spatial position, and a strong reliance on global contrast as a diagnostic shortcut. Both pointed toward the same solution: a richer feature representation trained with augmentation that specifically prevents over-reliance on any single image statistic, particularly contrast. This directly shaped the design of the ResNet50 pipeline described next.

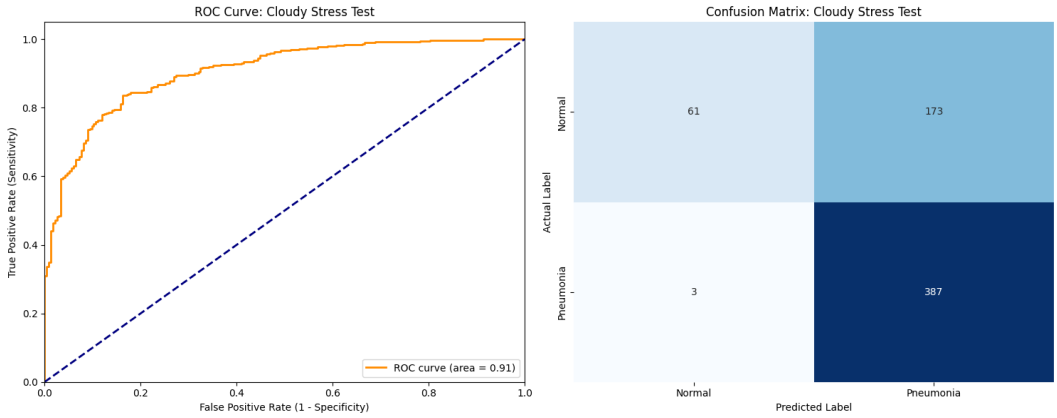


Figure 9: Cloudy Image Results.

6 ResNet50: Preprocessing, Architecture, and Results

6.1 Preprocessing

Building on the same balanced dataset from Section 3, the ResNet50 pipeline added two further preprocessing steps. First, images were dynamically upsampled from 128×128 to 224×224 during training to satisfy ResNet50's required input dimensions. Second, and crucially motivated by the stress test findings, on-the-fly data augmentation was applied to the training pipeline: random rotation (±5%), translation (±5%), zoom (±5%), and random contrast adjustment (±10%). The contrast augmentation was specifically chosen to prevent the model from over-fitting to the contrast statistics of the training set — directly addressing the shortcut behavior exposed by the cloudiness test. Augmentation was applied only during training; validation and test sets were processed without augmentation.

6.2 Architecture

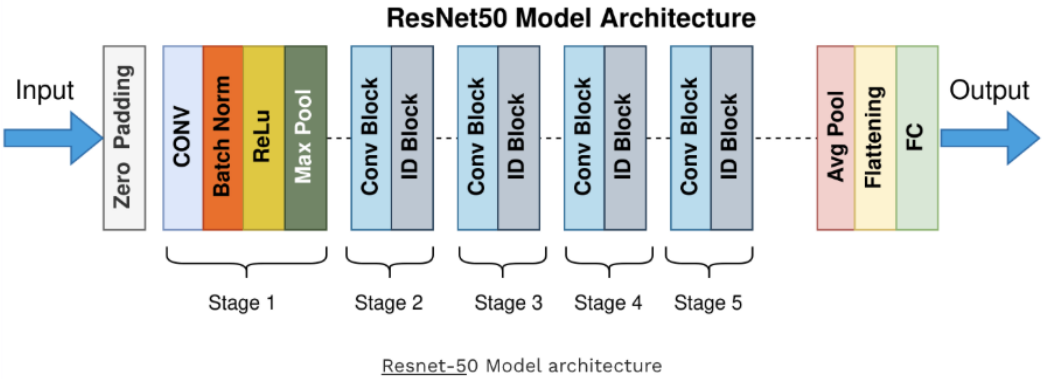


Figure 10: ResNet50 Model Architecture.

ResNet50 is a 50-layer deep residual network pre-trained on ImageNet's 1.2 million images. Its skip connections allow gradients to flow directly through shortcut pathways, effectively eliminating the vanishing gradient problem that prevented training of very deep networks. We employed a two-phase transfer learning strategy:

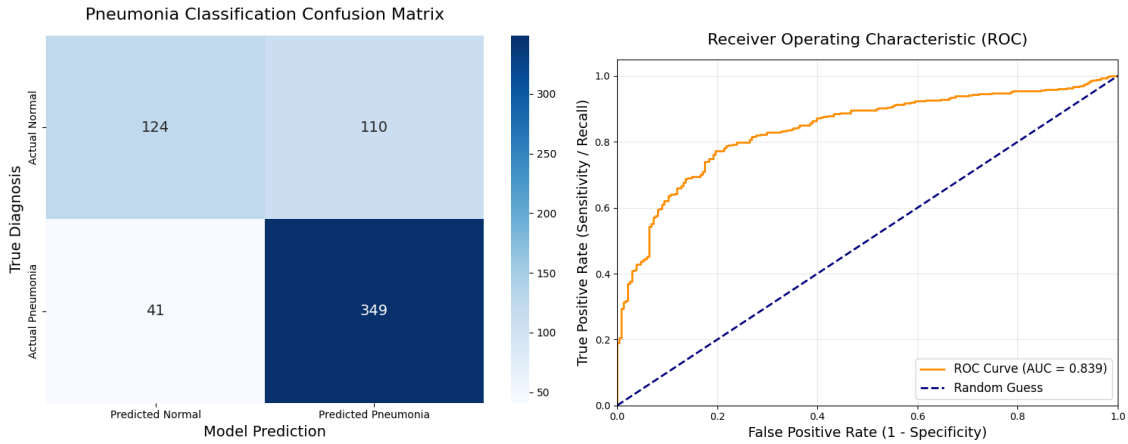
Phase 1 — Feature Extraction (Frozen Base): The entire ResNet50 base was loaded with pre-trained ImageNet weights and frozen. A custom classification head was appended: GlobalAveragePooling2D, Dense(128, ReLU), Dropout(0.5), and Dense(1, sigmoid). Only this head was trained (Adam, lr=1e-4) for up to 10 epochs with Early Stopping (patience=3). This leverages ImageNet's rich feature representations while training only the small head to remap them to our binary task.

Phase 2 — Fine-Tuning (Partial Unfreeze): Layers from index 143 onward were unfrozen while earlier layers remained frozen, allowing higher-level convolutional features to be gradually adapted to chest radiology while preserving the well-calibrated low-level detectors. The learning rate was reduced to 1e-5 to apply careful weight updates. A ReduceLROnPlateau callback halved the learning rate whenever validation loss stagnated for 2 consecutive epochs. The rationale for transfer learning here is practical: even after rebalancing, our training set of 2,294 images is far too small to train a 50-layer network from random initialization without severe overfitting. Transfer learning addresses this by initializing with pre-learned priors that generalize well with limited fine-tuning data.

The rationale for using ResNet50 over a simpler architecture was because the depth and skip connections allow the model to learn richer, more nuanced features than our shallow custom CNN. Also, our training dataset (even after rebalancing, 2,294 images) is small by deep learning standards, and is far too small to train a 50-layer network from random initialization without severe overfitting. Transfer learning directly addresses this by initializing the network with strong, pre-learned priors that generalize well even with limited fine-tuning data.

6.3 Results

After Phase 1 (frozen base), ResNet50 achieved 76% accuracy with more balanced class performance at 53% Normal recall compared to the custom CNN's 37%. After Phase 2 fine-tuning, the model reached its best performance on the test set:



(a) ResNet50 Confusion Matrix

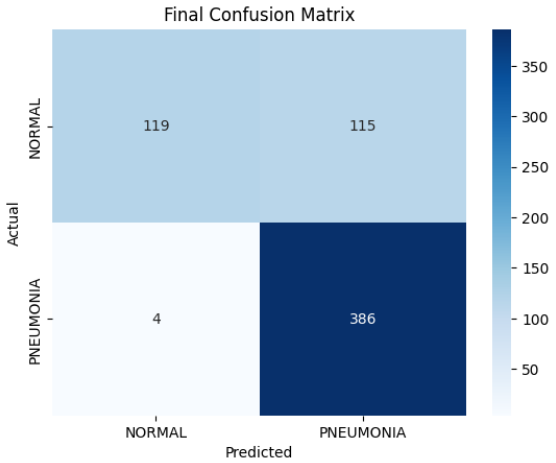
(b) ResNet50 ROC curve

Figure 11: Test Results for ResNet50 Base Model

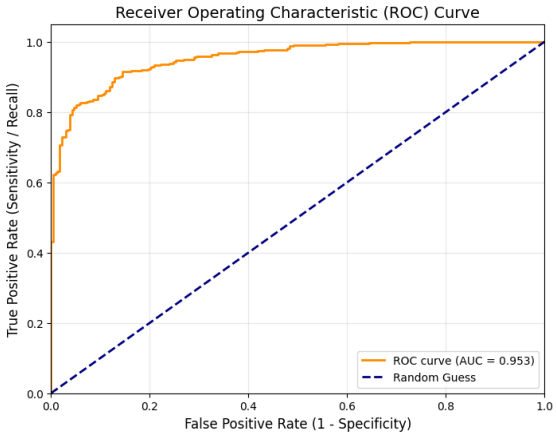
After Phase 2 fine-tuning, the ResNet50 model achieved the following results on the test set:

The fine-tuned ResNet50 achieved 80.93% overall accuracy with 77.05% precision and 98.97% pneumonia recall - correctly identifying 386 of the 390 true pneumonia cases. This represents a meaningful improvement over the custom CNN in both overall accuracy and Normal specificity, reflecting the combined benefits of richer feature representations, deeper architecture, and the contrast augmentation that prevented the shortcut-learning behavior exposed by the stress tests.

Given the medical nature of this classification task, we were particularly attentive to the asymmetric consequences of different error types. A false negative (a pneumonia case classified as normal) is far more dangerous clinically than a false positive (a normal case classified as pneumonia), because a missed diagnosis may delay life-saving treatment. Thus, we prioritized recall (true positive rate) as our primary metric alongside overall accuracy. Our model was able to successfully do this, and out of the 390 pneumonia patients, our model was able to correctly identify 382 of them.



(a) ResNet50 Fine Tuned Confusion Matrix



(b) ResNet50 Fine Tuned ROC curve

Figure 12: Test Results for ResNet50 Fine Tuned Model

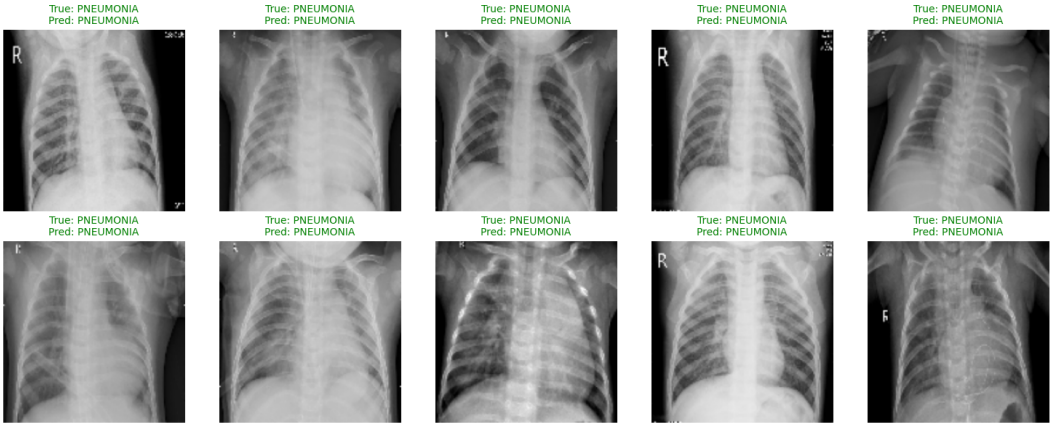


Figure 13: ResNet-50 Fine Tuned Model in Predicting Pneumonia.

7 Discussion

Our approach of building, testing, and redesigning the models was highly effective. The stress tests did more than just evaluate our first model; they exposed its specific flaws. Specifically, the cloudy image test revealed that the model was using overall image contrast as a diagnostic shortcut. While this trick worked on the original dataset, it failed completely when the images were altered. To fix this, we added random contrast adjustments to our ResNet training data. This forced the new model to learn true lung features, directly contributing to the improvement in Normal recall from 37% to roughly 55%.

Both models share a fundamental bias toward high Pneumonia recall (true positive rate) at the expense of Normal specificity (true negative rate). In a clinical context, high sensitivity (prioritizing the detection of all actual infections) is more desirable. Missing a pneumonia case is far more dangerous than a false alarm, but Normal recall of 37–55% is still too low for fully autonomous deployment. Adjusting the classification threshold above 0.5 would trade some Pneumonia recall for improved specificity, and calibrating this threshold using clinical cost estimates would be an important practical next step.

Both models' overall accuracy (76–81%) falls below published clinical benchmarks (>90%), attributable primarily to low input resolution (128×128 native, upsampled to 224×224 for ResNet), a relatively small training cohort, and simpler architectures compared to ensemble or attention-based approaches.

8 Conclusion

We built and evaluated two deep learning models for binary pneumonia classification from chest radiographs, following an intentional iterative process. The custom CNN (75.80% accuracy, 99.23% pneumonia recall) provided a strong baseline and, via stress testing, a clear failure mode: over-reliance on global image contrast as a diagnostic shortcut. These findings directly shaped the ResNet50 design, specifically the addition of contrast augmentation, which produced measurably more robust performance (80.93% accuracy, 98.97% recall) with better balanced sensitivity and specificity. Future work should focus on natively high-resolution inputs, Grad-CAM attribution to validate that model attention aligns with clinically relevant lung regions, and threshold calibration based on clinical cost-benefit analysis.

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